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Review

Influence of non-trapezoidal cross section on polarization characteristics and leakage loss of LNOI ridge waveguide

Ming Zhang[a], **Jing Zhang, Yu Fan, Zhaoqiong Yu, Ying Du**

College of Sciences, Zhejiang University of Technology, Hangzhou 310023, China

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Abstract In order to research the influence of waveguide cross-sectional shape on LNOI ridge waveguide, a non-trapezoidal shallow-etched ridge waveguide with an arc-shaped angle at the top of the ridge region was proposed. The polarization characteristics and leakage loss of the waveguide under different cross-sectional parameters as well as the relationship between leakage loss and transmittivity under different angles of sidewall were analyzed using vector finite element method. It is found that the smaller the leakage loss, the greater the transmittivity. Additionally, increasing the radius of arc-shaped angle and the angle of sidewall significantly enhances the polarization properties and reduces leakage loss and appropriate arc-shaped angle can facilitate mode conversion, thereby optimizing waveguide performance. Therefore, the results are expected to provide valuable references for further optimizing the design and fabrication of LNOI ridge waveguide.

1 Introduction

LNOI is a material prepared by bonding sub-micrometer-thick lithium niobate films to a SiO_2 substrate [1, 2]. It inherits the excellent electro-optic, acousto-optic, and nonlinear properties of LN crystals, and it has a good prospect of application in integrated optical devices [3–5]. In addition, LNOI ridge waveguide has a high refractive index contrast for strong confinement and low-loss transmission of light [6]. In waveguide, loss is an important parameter for evaluating the performance of waveguide, and the loss of ridge waveguide is mainly caused by mode conversion at the ridge boundary [7]. In waveguides, the ridge height, ridge width, and trench width are all important parameters that affect the polarization characteristics and leakage loss.

The shape of the waveguide cross section is one of the factors that affect the polarization characteristics and leakage loss of ridge waveguide, and relevant studies have been carried out in recent years. For example, in 2007, Webster [8] et al. obtained a shallow-etched ridge waveguide with a non-rectangular cross section after using thermal oxygen method to treat SOI ridge waveguide and firstly observed unusual cyclic leakage mode of TM-like mode in this waveguide with different waveguide widths. Due to the difficulty in manufacturing waveguides without sidewalls, the influence of angled sidewalls on mode conversion is studied in 2021 [9]. Later, Yao [10] et al. theoretically analyzed the effect of curved cross section on non-rectangular SOI ridge waveguide. E. Saitoh [11] et al. studied the polarization properties and leakage loss of trapezoidal cross-sectional LNOI ridge waveguide by the vector finite element method theory. The study showed that the Z-cut LNOI ridge waveguide exhibits a leakage loss behavior similar to that of the SOI ridge waveguide, while in the X-cut LNOI ridge waveguide, the height of the ridge waveguide affects the polarization properties and the leakage loss of the waveguide; this causes the leakage mode to convert between TE mode and TM mode. From the literature [8] and [10], it can be observed that the SOI waveguide cross section exhibits a rectangular shape, and at its upper top corner, there exists an arc-shaped angle instead of the strict rectangular shape as designed. At present, there are no relevant studies exploring the effect of this cross-sectional shape on trapezoidal LNOI waveguide. In view of this, how this will affect the polarization characteristics and leakage loss of LNOI ridge waveguide is a topic worth discussing.

Based on the above, in order to explore the effect of the presence of arc-shaped angle on the LNOI ridge waveguide, this article proposes a waveguide with a non-trapezoidal cross section by changing the top corner of the trapezoidal cross section of the ridge waveguide to an arc-shaped angle and simulates it using vector finite element method. Firstly, we analyzed the effects of different trench widths and ridge widths on non-trapezoidal LNOI ridge waveguide. Then, the relationship between transmittivity and leakage loss under different sidewall angles is studied. Finally, we compared our proposed non-trapezoidal LNOI ridge waveguide with strict trapezoidal ridge waveguide. During this process, the leakage loss and transmittivity of the waveguide in TE-like mode and TM-like mode under X-cut and Z-cut conditions are observed, respectively. By analyzing the simulation results, it is found that the variation of the arc-shaped angle can significantly affect the polarization characteristics and leakage loss of the LNOI ridge waveguide.

[a] e-mail: cim2046@zjut.edu.cn (corresponding author)

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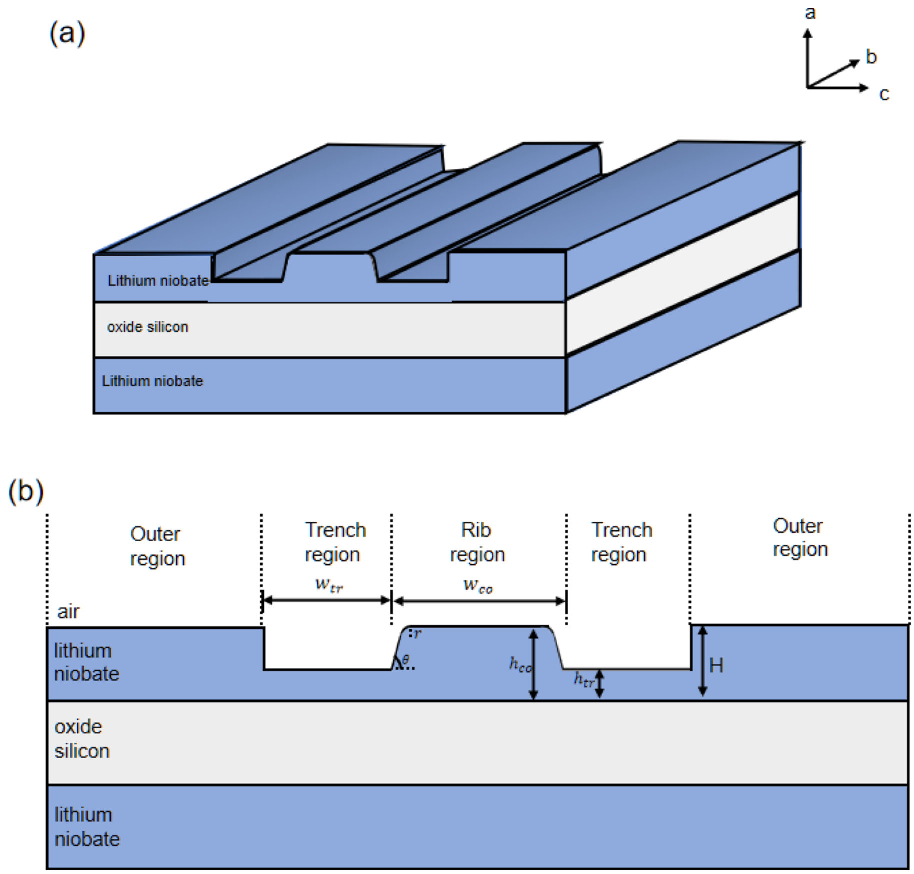
Fig. 1 Schematic of non-trapezoidal LNOI ridge waveguide



Larger arc radius and sidewall angle can reduce the leakage loss of the waveguide and make the waveguide have better polarization characteristics. In addition, it can be found that selecting the appropriate radius of the arc and sidewall angle can achieve mode conversion of the waveguide, thereby further improving the application potential of the waveguide.

2 Waveguide model and theoretical analysis

2.1 Theoretical model of non-trapezoidal cross-sectional LNOI ridge waveguide

Our proposed non-trapezoidal LNOI ridge waveguide is shown in Fig. 1. In the figure, SiO_2 is used as the substrate and its thickness is kept as 1 µm. The entire waveguide layer from left to right is the outer region (boundary layer), trench region, rib region (waveguide layer), trench region, and outer region. In the whole model, the thickness of waveguide is H, its width is W, and its length is L. Its corresponding material is lithium niobate.

Lithium niobate is a kind of anisotropic material, and the refractive index distribution is related to the direction of crystal cutting. According to the direction of the crystal axis relative to the wafer, it can be divided into Z-cut and X-cut [12]. The orientation of the crystal is represented by a, b, c in the Fig. 1a. When the LN is Z-cut, a $= n_e$ and b $=$ c $= n_o$; and when the LN is X-cut, a $=$ b $= n_o$, c $= -n_e$, and '−' represents the negative direction. Figure 1b shows the cross section of the waveguide. The width and thickness of the trench are w_{tr} and h_{tr}, respectively; the width and thickness of the ridge are w_{co} and h_{co}, respectively; the angle of sidewall is θ; and the radius of the etched arc-shaped angle at the upper corner is r. And the parameters set in the model are: $H = 1$ µm, $W = 20$ µm, $L = 10$ µm, $h_{co} = 1$ µm, $h_{tr} = 0.82$ µm, incident wave $\lambda = 1.55$ µm, $n_1 = n_{SiO2} = 1.444$, $n_2 = n_{\mathrm{air}} = 1$, effective refractive index of lithium niobate $n_o = 2.211$, and $n_e = 2.138$.

2.2 Theoretical analysis of mode conversion and leakage loss

There are usually two types of guided modes in ridge waveguide, which are divided into ridge mode and slab mode based on the region restricted by the mode [13]. When light propagates to the boundary in the ridge, due to the discontinuity at the ridge boundary, some waves will produce mode conversion and leak into the trench. If this mode disappears quickly, the guided mode loss in the

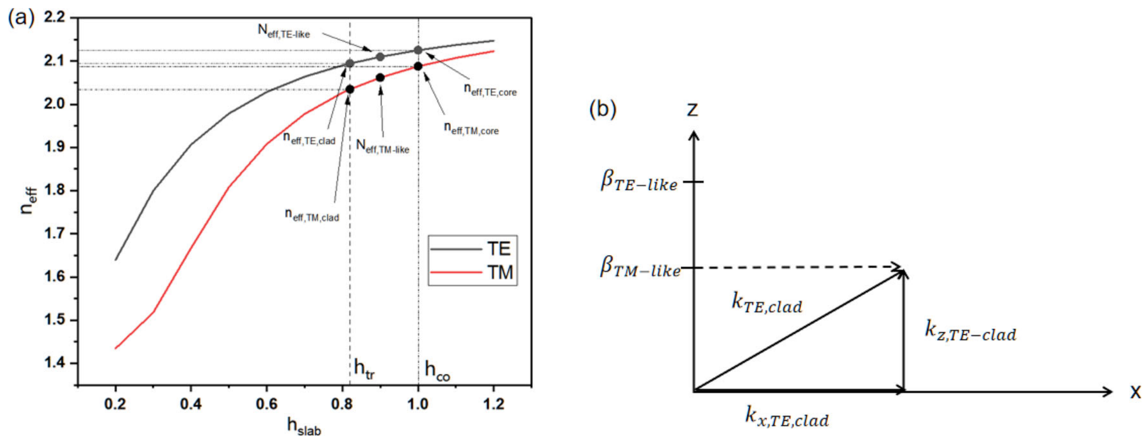


Fig. 2 **a** Variation of effective refractive index of TE and TM mode of Z-cut LNOI slab waveguide with the thickness of slab. **b** Schematic diagram of mode conversion phase matching

ridge region will remain at an extremely low level; if this mode continues to propagate in the trench region, the guided mode in the ridge region will generate inherent leakage losses due to mode conversion.

Here we use Z-cut LNOI slab waveguide as an example to analyze whether the waveguide will produce mode conversion. Figure 2a shows the variation of effective refractive index of TE and TM mode in Z-cut LNOI slab waveguide with slab thickness h_{slab}. In the figure, $n_{\text{eff, TE, core}}$, $n_{\text{eff, TE, clad}}$ and $n_{\text{eff, TM, core}}$, $n_{\text{eff, TM, clad}}$ represent the effective refractive indices of TE and TM mode in the ridge and trench regions of waveguide, respectively.$N_{\text{eff, TE}-\text{like}}$ and $N_{\text{eff, TM}-\text{like}}$, respectively, represent the effective refractive index of TE-like mode and TM-like mode. In addition, $\beta_{\text{TE}-\text{like}}$ and $\beta_{\text{TM}-\text{like}}$ represent the propagation constants of TE-like and TM-like mode in the ridge and trench regions, respectively, while $k_{\text{TE, clad}}$ represents the propagation constants of TE mode in the trench region.

Figure 2a shows that the effective refractive index of TE-like mode is always higher than that of TM-like mode, because the propagation constant $\beta = k_0 N_{\text{eff}}$, which means that the propagation constant of TE-like mode is always larger than that of TM-like mode. And the effective refractive index of TE-like mode and TM-like mode is between $n_{\text{eff, TE, core}}$, $n_{\text{eff, TE, clad}}$ and $n_{\text{eff, TM, core}}$, $n_{\text{eff, TM, clad}}$, respectively. The mode conversion at the ridge boundary is always polarized from a low effective refractive index to a high effective refractive index. For the TE-like mode, because the effective refractive index $N_{\text{eff, TE}-\text{like}}$ greater than $n_{\text{eff, TE, clad}}$ and $N_{\text{eff, TM, clad}}$, the mode at the edge of the ridge that leaks into the trench region due to mode conversion will quickly disappear, and the impact on leakage loss is relatively small; for TM-like mode, because $N_{\text{eff, TM}-\text{like}}$ is greater than $n_{\text{eff, TM, clad}}$ and less than $n_{\text{eff, TE, clad}}$, there is $k_{\text{TE, clad}} > \beta_{\text{TM}-\text{like}}$. At this point, there will be an angle that causes phase matching between $\beta_{\text{TM}-\text{like}}$ and $k_{\text{TE, clad}}$ in the Z direction, resulting the TM–TE mode conversion of the ridge waveguide. That is to say, the conversion of mode into guided mode in the trench region can lead to leakage losses. Figure 2b shows a schematic diagram of the corresponding mode conversion phase matching.

The same principle can be used to analyze the mode conversion of the X-cut LNOI slab waveguide, and Fig. 3 shows the variation of the effective refractive index of TE and TM mode in X-cut LNOI slab waveguide with the thickness of the slab h_{slab}. From the figure, it can be seen that the mode conversion in X-cut LNOI slab waveguide is more complicated. When the thickness of the slab is 0.72 μm, the effective refractive indices of TE and TM mode are the same. It is expected that there will be no mode conversion here. When the h_{co} and h_{tr} are thinner than 0.72 μm, the effective refractive index of TM-like mode in the ridge is lower than that of TE mode in the trench, which may result in leakage; and when the thickness of h_{co} and h_{tr} is larger than 0.72 μm, the effective refractive index of TM mode in the trench is higher than that of TE-like mode in the ridge, which may result in leakage loss of TE-like mode.

The behavior of leakage loss can be explained by the theory of interference [14], Fig. 4 shows the theoretical model of the waveguide leakage loss when it reaches its peak value, the mode conversion between the TM-like mode in the ridge region and the TE higher-order mode in the trench region was demonstrated and the condition when the destructive interference is satisfied in the trench is shown in Eq. (1):

$$2w_{\text{tr}}k_{x,\text{TE,clad}} + \varnothing_1 + \varnothing_2 - k_{x,\text{TM}-\text{like}}r = 2\ \text{m}\pi \tag{1}$$

where w_{tr} is the width of the trench region, $k_{x,\text{TE, clad}}$ is x-component of the wave vector of the TE mode in the trench, $k_{x,\text{TM}-\text{like}}$ is x-component of the wave vector of the TM-like mode in the ridge region, and $\varnothing_1$ and $\varnothing_2$ are the phase differences introduced during mode conversion and transmission at the ridge boundary. # 1 is the interface between the ridge region and the trench, and # 2 is the interface between the trench region and the outer layer. The r in Eq. (1) is the distance from the equiphase point of TM-like mode to the boundary #1, as shown in Fig. 4. When the parameters and cutting direction in the LNOI are fixed, all of the above parameters are quantitative. When m in Eq. (1) is odd, the coherent subtraction will be produced in the trench region and the energy

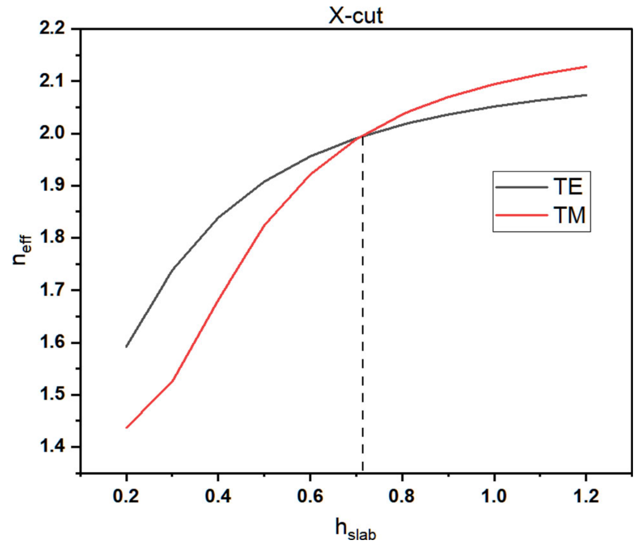


Fig. 3 Variation of effective refractive index of TE mode and TM mode of X-cut LNOI slab waveguide with the thickness of slab h_{slab}

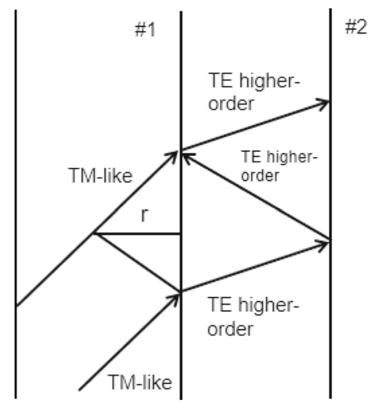


Fig. 4 Schematic of mode conversion

is limited to the ridge region, which results in lower leakage loss and increased transmission in the LNOI for TE_0 and TM_0 mode. Therefore, by changing the width of the trench, the interference of light incident from the guided wave layer to the trench region can be adjusted.

In addition, the distance between the two resonance points, which is the size of the resonance period Δw_{tr}, is shown in Eq. (2):

$$\Delta w\text{tr} = \frac{\lambda}{2\sqrt{n_{eff,clad}^2 - N_{eff}^2}} \tag{2}$$

When TM-like mode is leaky, $n_{\text{eff, clad}} = n_{\text{eff, TE, clad}}$, $N_{\text{eff}} = N_{\text{eff, TM-like}}$; and when TE-like mode is leaky, $n_{\text{eff, clad}} = n_{\text{eff, TM, clad}}$, $N_{\text{eff}} = N_{\text{eff, TE-like}}$.

By calculating the complex propagation constant β of ridge mode, leakage loss can be obtained. And β can be obtained by solving the matrix eigenvalue equation under the perfectly matched layer (PML), and then, the leakage loss can be obtained through Eq. (3) [15]:

$$\text{Leakage loss} = 8.686\,\text{Im}\{\beta\} \tag{3}$$

3 Analysis of results

3.1 The effect of trench width on polarization characteristics and leakage loss of Z-cut non-trapezoidal LNOI ridge waveguide

Figure 5 shows the variation of leakage loss and effective refractive index with trench width for TE-like mode, TM-like mode, and TE higher-order mode in the Z-cut non-trapezoidal LNOI ridge waveguide for the case of $w_{co} = 1$ μm, $\theta = 40^{\circ}$, $r = 0.1$ μm. Figure 5a and b shows that the effective refractive index of TE-like mode changes relatively weakly with the increase in the trench width, and its value floats between 2.0998 and 2.1002. This is because the waveguide has a strong constraint on the TE-like mode, so that the TE-like mode leakage loss decreases rapidly and maintains at a low level. However, the variation of the effective refractive index and leakage loss of the TM-like mode and the TE higher-order mode are more complex, and it exhibits an obvious periodicity,

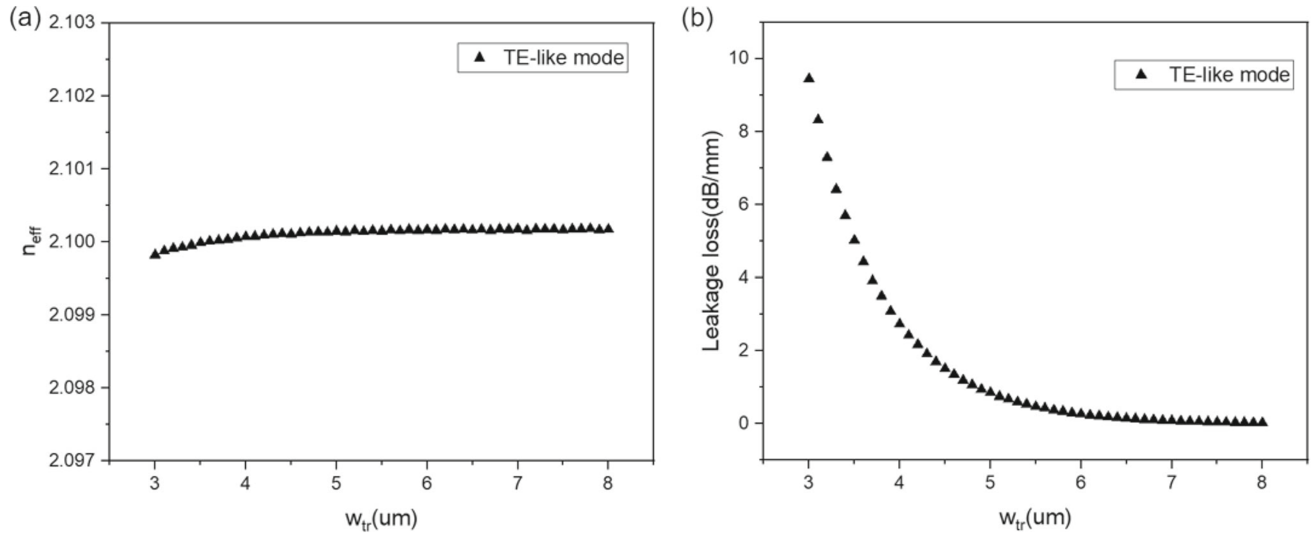


Fig. 5 Variation of **a** effective refractive index and **b** leakage loss with trench width for TE-like mode in Z-cut non-trapezoidal LNOI ridge waveguide

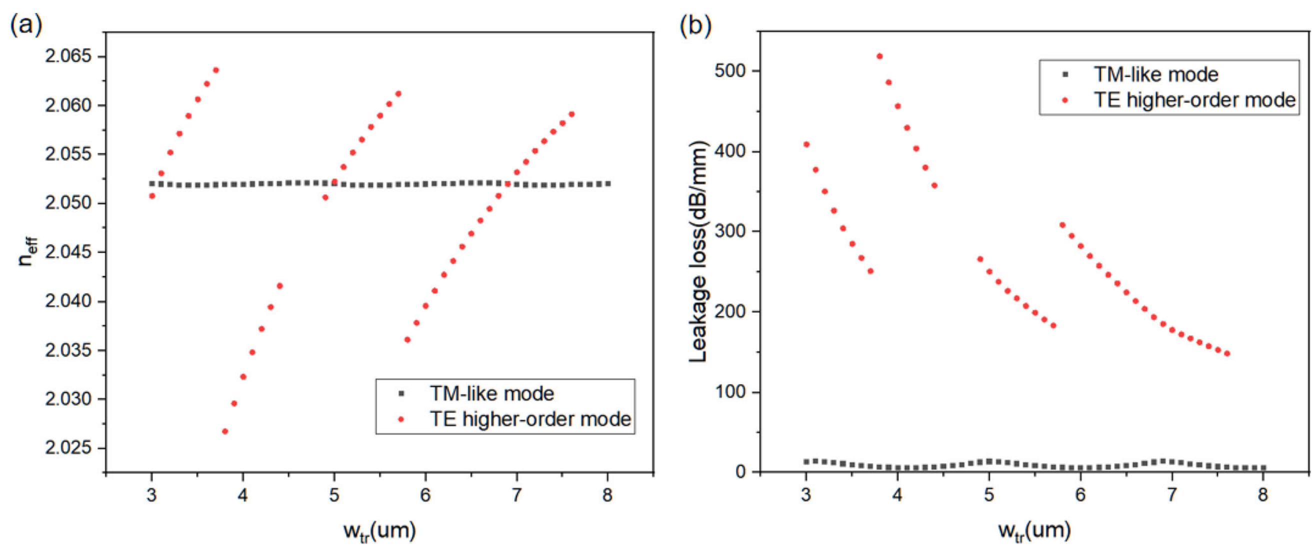


Fig. 6 Variation of **a** effective refractive index and **b** leakage loss with trench width for TM-like mode and TE higher-order mode in Z-cut non-trapezoidal LNOI ridge waveguide

which is mainly due to the mode conversion at the ridge boundary, which can also be explained by the mode coupling. Figure 6a and b shows that the effective refractive index of TE higher-order mode and TM-like mode has three intersections at $w_{tr} = 3.1$ µm, 5 µm, and 6.9 µm, and at the intersections, the leakage loss of the TM-like mode reach the maximum value.

Figures 7, 8, and 9 show the electric field components of TM-like mode and TE higher-order mode at $w_{tr} = 5$ µm, 6 µm, and 6.9 µm, respectively. From the electric field distribution diagram, it can be seen that the mode distribution of TM-like mode is very similar to that of TE higher-order mode, and as the w_{tr} increases, TM-like mode will couple with higher-order mode. At $w_{tr} = 5$ µm, the TM-like mode in the ridge region will couple with the fifth-order TE mode in the trench region; at $wtr = 6.9$ µm, the TM-like mode is coupled with the seventh-order TE mode. And in these two points, the E_x and E_y electric field components of TM-like mode and TE higher-order mode are very similar. At this time, the TM/TE mode conversion reaches its maximum, at which point the TM-like mode leakage loss reaches its peak. However, at $w_{tr} = 6$ µm, which is the minimum leakage loss of TM-like mode, there is a significant difference in the mode diagrams between TE higher-order mode and TM-like mode. The E_y center of TE higher-order mode is obvious, while E_x is weaker near the center and stronger away from the center. But at this time, both E_x and E_y of TM-like mode are more obvious, and the TM/TE conversion is the smallest. Therefore, the leakage loss of TM-like mode is the smallest.

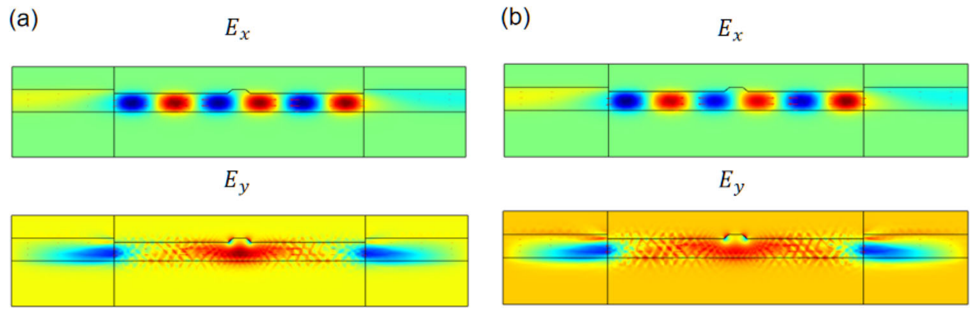


Fig. 7 **a** TM-like mode and **b** TE higher-order mode electric field components at $w\mathrm{tr} = 5\ \mu\mathrm{m}$

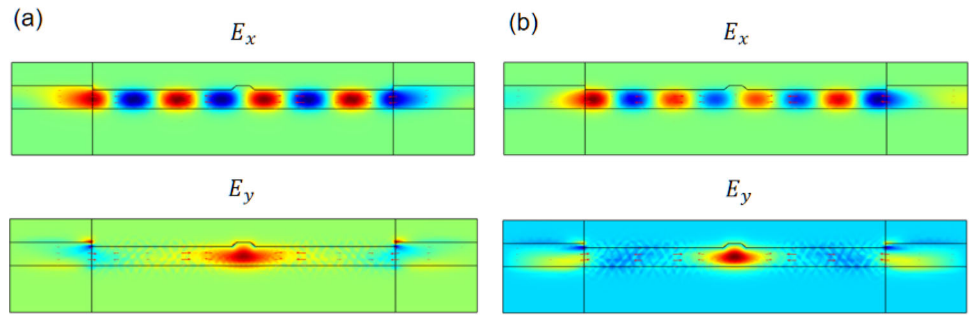


Fig. 8 **a** TM-like mode and **b** TE higher-order mode electric field components at $w\mathrm{tr} = 6\ \mu\mathrm{m}$

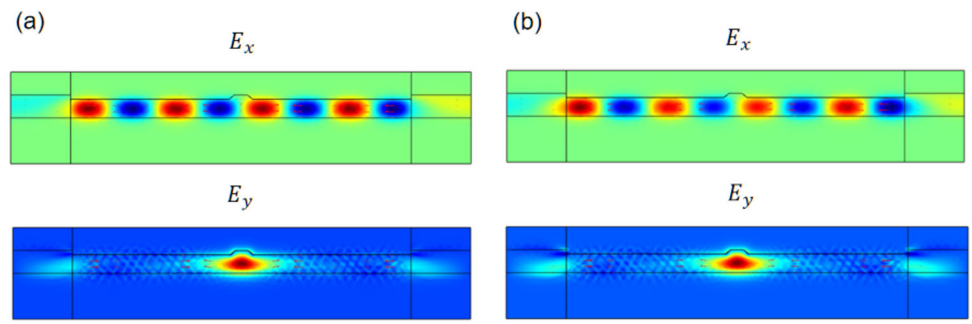


Fig. 9 **a** TM-like mode and **b** TE higher-order mode electric field components at $w\mathrm{tr} = 6.9\ \mu\mathrm{m}$

3.2 The effect of ridge width on polarization characteristics and leakage loss of non-trapezoidal LNOI ridge waveguide

Figures 10 and 11 simulate the leakage loss and effective refractive index of TE-like and TM-like mode in Z-cut LNOI ridge waveguide with different ridge width. It can be observed that the leakage loss of TM-like mode exhibits periodic variations. According to Eq. (2), as w_{co} and w_{tr} increase, the leakage loss of TM-like mode increases. This is mainly because that $N_{\mathrm{eff,\,TM-like}}$ increases and $n_{\mathrm{eff,\,TE,\,clad}}$ decreases, causing the resonance point to move to the right and the leakage loss to increase.

Figure 12 shows the leakage loss of the X-cut LNOI non-trapezoidal cross-sectional waveguide at different w_{co} under the same parameters as Z-cut. It can be observed that with the increase of w_{tr} and w_{co}, the leakage loss of TM-like mode will rapidly decrease and maintain at an extremely low level, similar to the TE-like mode of Z-cut non-trapezoidal LNOI ridge waveguide; the leakage loss of TE-like mode increases with the increase of w_{co} and w_{tr}, similar to the TM-like mode of Z-cut non-trapezoidal LNOI ridge waveguide. When $w_{\mathrm{co}} = 1.4\ \mu\mathrm{m}$ and $w_{\mathrm{tr}} = 6\ \mu\mathrm{m}$, the leakage loss in TE-like mode is 23.9 dB/mm, and the leakage loss in TM-like mode is 0.00012 dB/mm, which has sufficient extinction ratio. Based on this principle, polarizers can be designed to achieve higher extinction ratios by selecting sizes with wider w_{co} and longer w_{tr}.

3.3 The relationship between the transmittivity and leakage loss under different sidewall angles in non-trapezoidal LNOI ridge waveguide

In the previous section, the different performances of effective refractive index and leakage loss in TE-like mode and TM-like mode with varying trench width and ridge width are studied. Here we want to verify the relationship between leakage loss and transmittivity. The method used here is to establish a COMSOL three-dimensional non-trapezoidal LNOI ridge waveguide and then observe the relationship between the transmittivity and leakage loss of TE-like mode and TM-like mode under different sidewall angle effects. Here, $w_{\mathrm{co}} = 1\ \mu\mathrm{m}$, $w_{\mathrm{tr}} = 6\ \mu\mathrm{m}$, and $r = 0.1\ \mu\mathrm{m}$.

Through verification, it can be seen from the results in Fig. 13 that for Z-cut LNOI non-trapezoidal ridge waveguide, the angle of the sidewall has an impact on both their transmittivity and leakage loss. As the result shows, with the increase in the sidewall

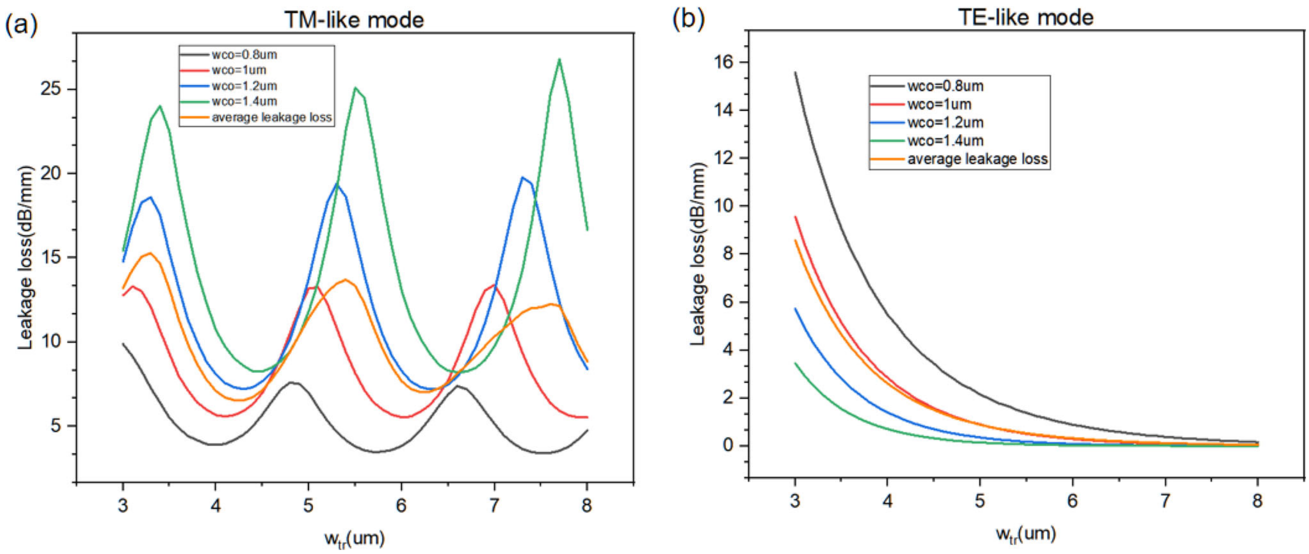


Fig. 10 Variation in leakage loss of **a** TM-like and **b** TE-like mode with ridge width in Z-cut non-trapezoidal LNOI ridge waveguide

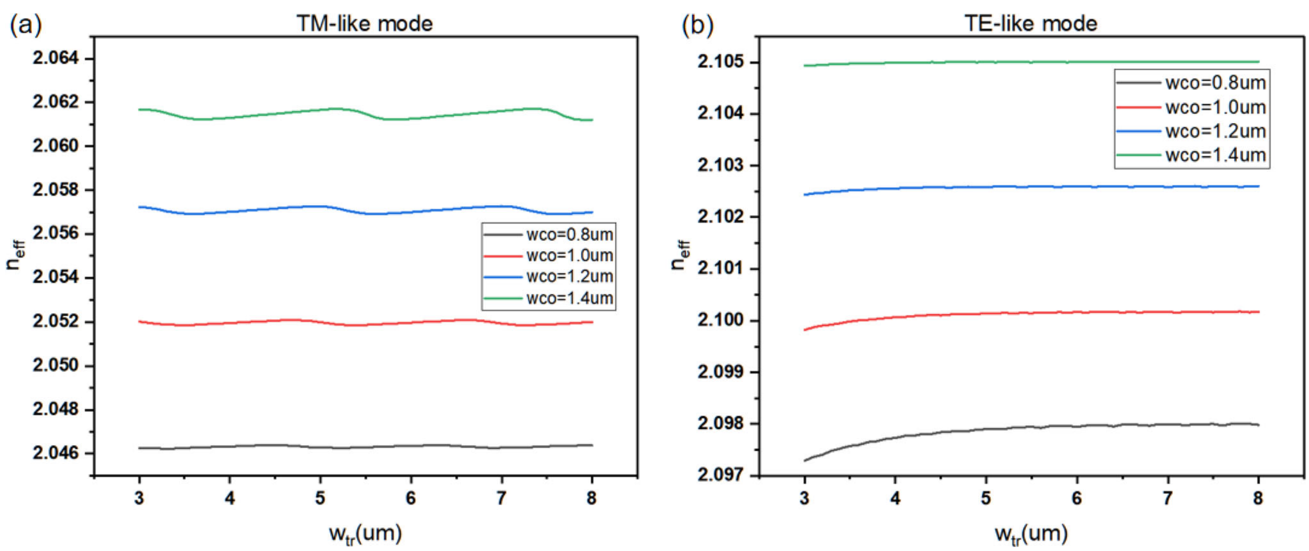


Fig. 11 Variation in effective refractive index of **a** TM-like and **b** TE-like mode with ridge width in Z-cut non-trapezoidal LNOI ridge waveguide

angle, the leakage loss of TM-like gradually increases; but it has little effect on the TE-like mode, almost no effect, and the larger the angle, the smaller the leakage loss. The transmittivity of the TE-like mode is increasing and reaches its peak at a sidewall angle of 80°, with a transmittivity of 0.99987; for the TM-like mode, it is the opposite situation, as the angle increases, the leakage loss gradually increases, and the transmittivity gradually decreases. Moreover, the transmittivity of TE-like mode is higher than that of TM-like mode.

For X-cut LNOI non-trapezoidal waveguide, the angle also affects the waveguide transmittivity and leakage loss. Figure 14 shows that in the X-cut TE-like mode, with the angle increasing, the leakage loss gradually increases and the transmittivity gradually decreases; in the X-cut TM-like mode, leakage loss also decreases with increasing angle, the main reason is that as the sidewall angle increases, the confinement effect of the waveguide core layer on light may become more significant, and the transmittivity decreases obviously once at the 65°, perhaps the distribution of the light field here has changed, and then, it still has an increasing trend, and this may be due to the excessive angle of the sidewall weakening the constraint ability of the ridge area on light rays. However, the transmittivity level of the TM-like mode is obviously lower than that of the TE-like mode.

Therefore, it can be concluded that there is a relationship between the transmittivity and leakage loss of non-trapezoidal waveguide: The smaller the leakage loss, the greater the transmittivity.

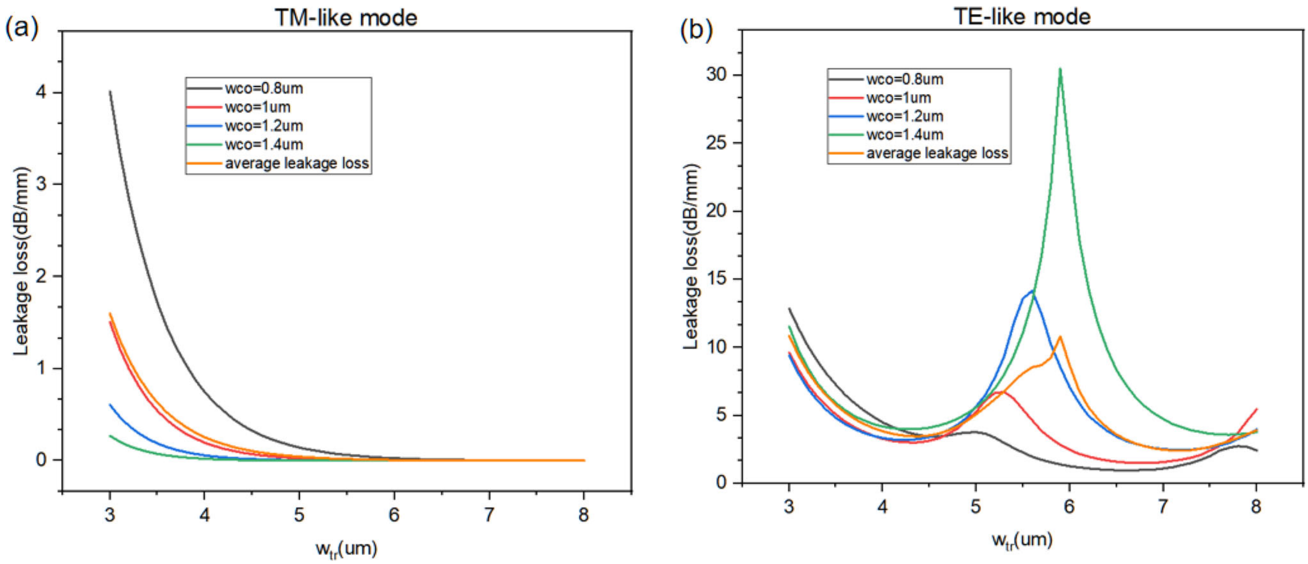


Fig. 12 Variation in leakage loss of **a** TM-like and **b** TE-like mode with ridge width in X-cut non-trapezoidal LNOI ridge waveguide

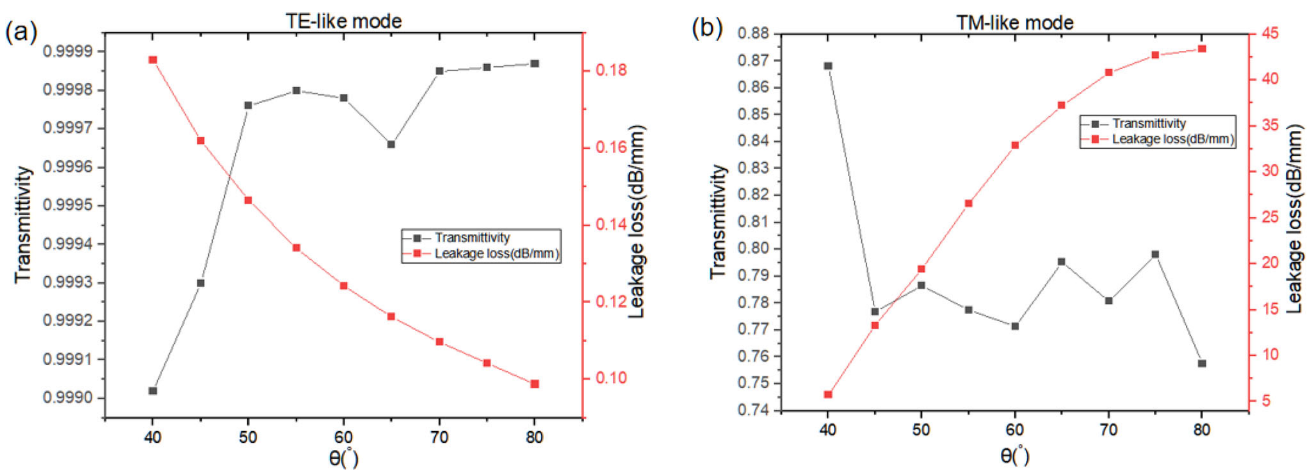


Fig. 13 Variation of **a** TE-like mode, **b** TM-like mode transmittivity and leakage loss with sidewall angle in Z-cut non-trapezoidal LNOI ridge waveguide

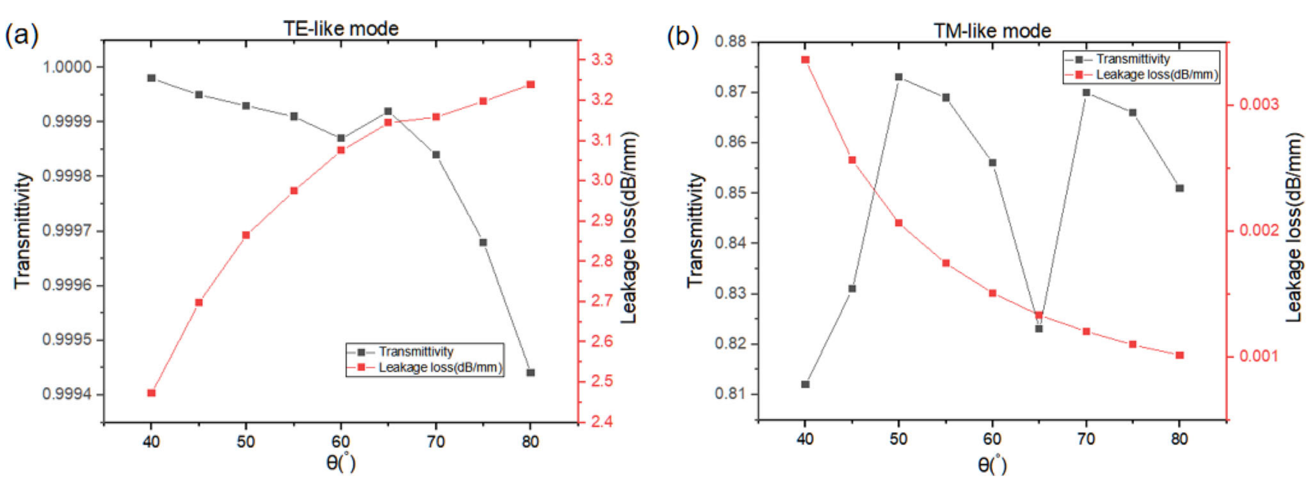


Fig. 14 Variation of **a** TE-like mode, **b** TM-like mode transmittivity and leakage loss with sidewall angle in X-cut non-trapezoidal LNOI ridge waveguide

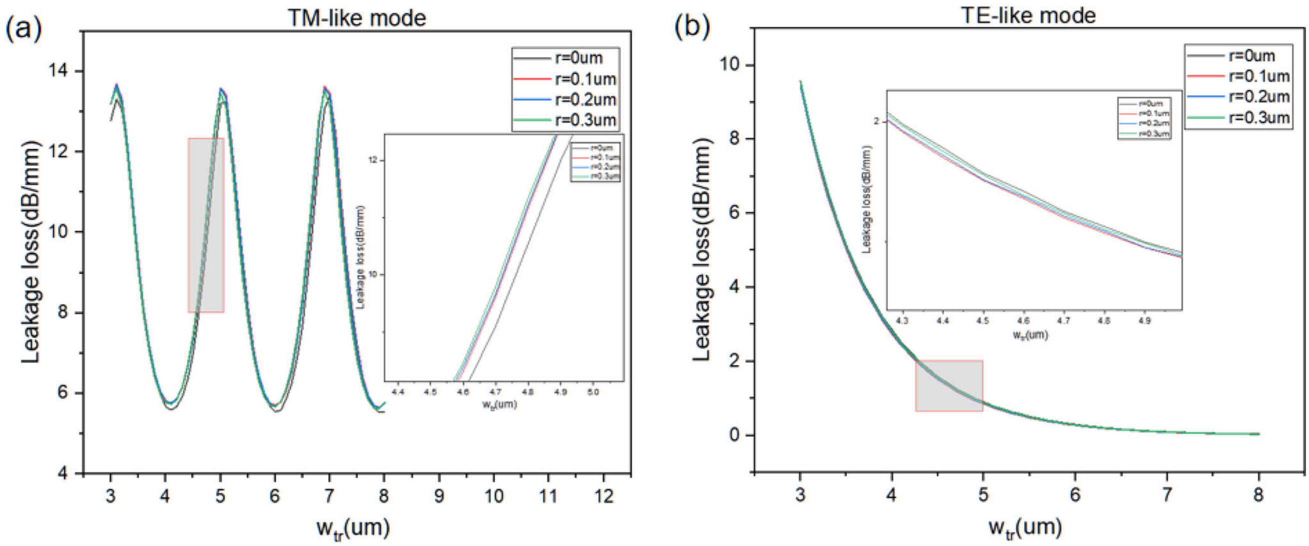


Fig. 15 Variation in leakage loss of **a** TM-like and **b** TE-like mode with radius of arc-shaped angle in Z-cut non-trapezoidal LNOI ridge waveguide

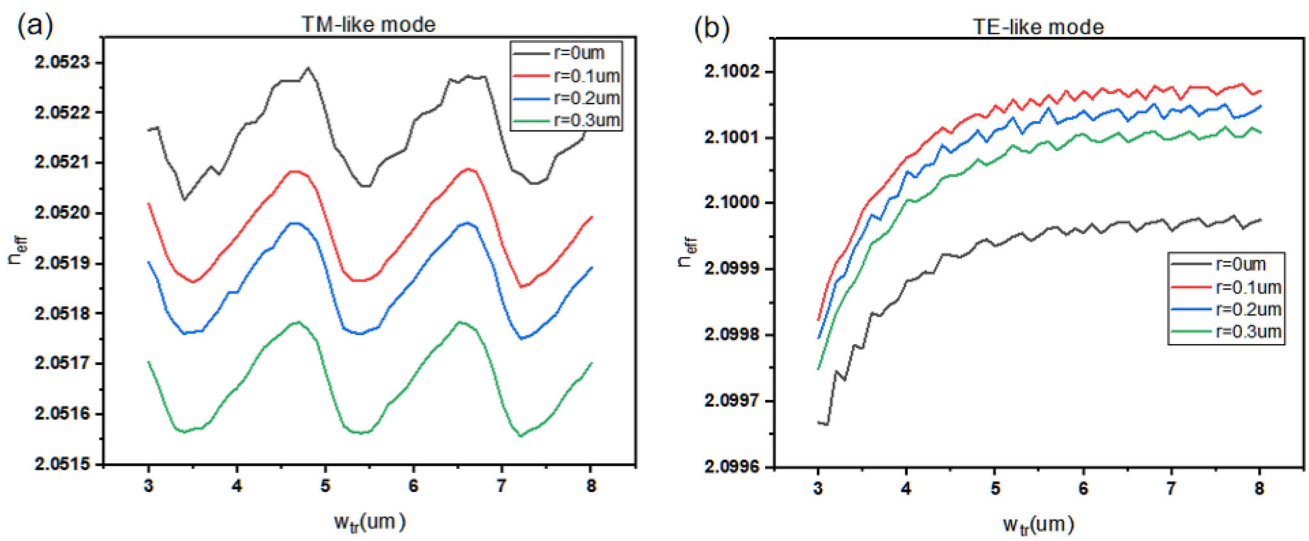


Fig. 16 Variation in effective refractive index of **a** TM-like and **b** TE-like mode with radius of arc-shaped angle in Z-cut LNOI ridge waveguide

3.4 Comparison between non-trapezoidal and trapezoidal LNOI ridge waveguide

Here, we compared the difference between Z-cut and X-cut LNOI non-trapezoidal cross-sectional ridge waveguide and ordinary trapezoidal cross-sectional waveguide, as shown in Figs. 15, 16, 17, and 18 for the case of $w_{co} = 1\ \mu$m, $\theta = 40^{\circ}$. Considering the ridge waveguide with ordinary trapezoidal cross section as a special case when $r = 0\ \mu$m, it can be observed that as the radius of the arc angle r increases, the change in the effective refractive index of the TE-like and TM-like mode is relatively weak. Compared with ordinary trapezoidal sectional ridge waveguide, the presence of arc-shaped angles increases the leakage loss of TE-like and TM-like mode to a certain extent, but the extinction ratio increases, and the resonance point position is not affected. It is easy to find that leakage losses of both TE-like and TM-like are low in X-cut LNOI waveguide; in Z-cut LNOI waveguide, at $w_{tr} = 6\ \mu$m, $r = 0.1\ \mu$m, 0.2 μm, and 0.3 μm, the leakage losses of TM-like mode are around 5.707 dB/mm, 5.678 dB/mm, and 5.668 dB/mm, respectively. It can be seen that the waveguide loss is not sensitive to changes in the radius of arc angle, thus improving the tolerance of the fabrication.

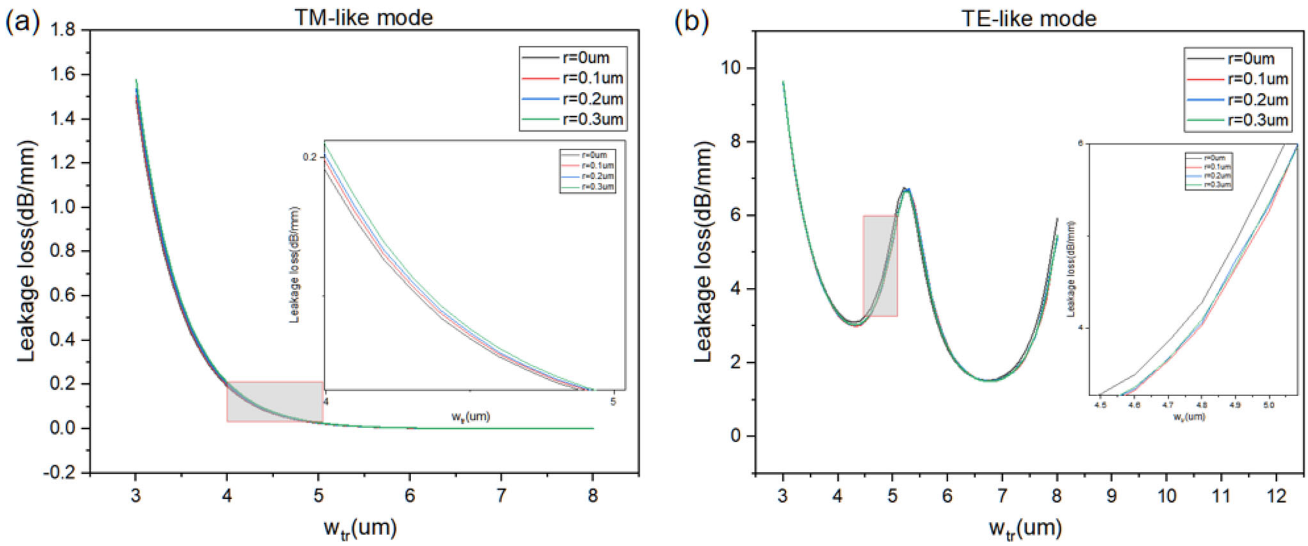


Fig. 17 Variation in leakage loss of **a** TM-like and **b** TE-like mode with radius of arc-shaped angle in X-cut non-trapezoidal LNOI ridge waveguide

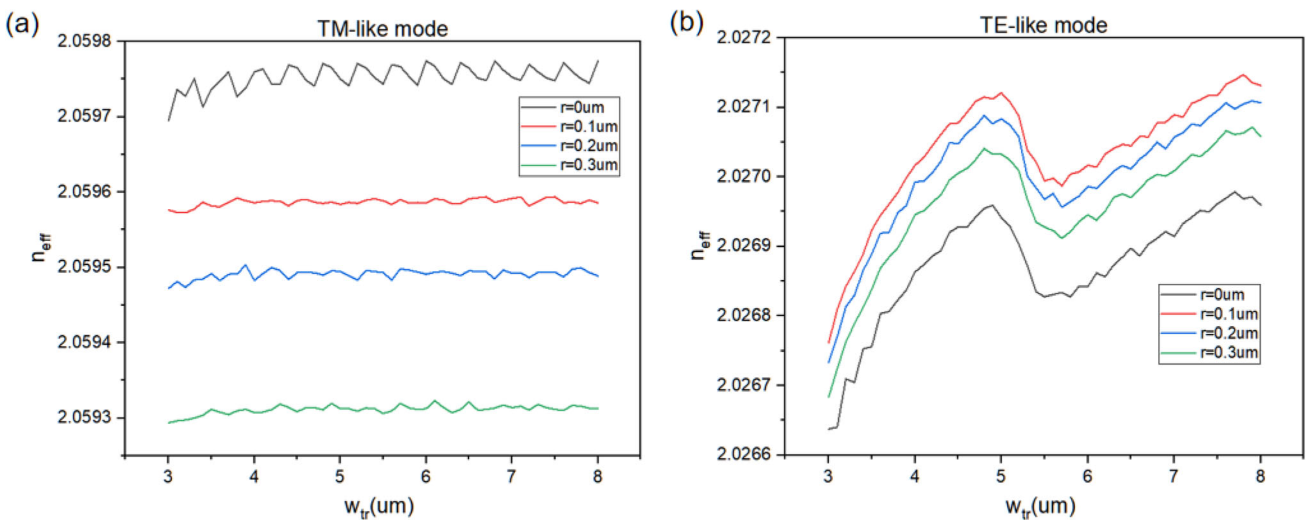


Fig. 18 Variation in effective refractive index of **a** TM-like and **b** TE-like mode with radius of arc-shaped angle in X-cut non-trapezoidal LNOI ridge waveguide

4 Discussion

From the full article, it can be seen that when the sidewall angle $\theta = 40°$, the radius of arc-shaped angle $r = 0.1$ µm, the ridge width $w_{co} = 1$ µm, and the trench width $w_{tr} = 6$ µm, the transmittivity of the TM-like mode in the Z-cut non-trapezoidal LNOI waveguide is the highest and the leakage loss is the lowest; at this point, the TE-like mode in the X-cut non-trapezoidal LNOI waveguide also has the highest transmittivity and the lowest leakage loss. And the leakage losses of Z-cut TE-like and X-cut TM-like modes are both very small. However, the leakage loss of the waveguide is lower when $w_{co} = 0.8$ µm. Due to the fact that we only simulated the effects of parameters such as ridge width and trench width on the leakage loss of non-trapezoidal LNOI waveguide when the radius of the arc angle $r = 0.1$ µm, and the arc angle radius had a relatively small impact on the leakage loss of LNOI ridge waveguide, we only found the optimum point for all parameters except for the ridge width w_{co}.

5 Conclusion

This article proposes a non-trapezoidal cross-sectional LNOI ridge waveguide and investigates the variation of effective refractive index, leakage loss, and transmittivity of the waveguide under different cross-sectional parameters by using vector finite element method to limit the perfect matching layer under frequency domain condition.

Through research, we found that: (1) leakage loss and transmittivity can be increased or decreased by controlling the angle of sidewall and radius of arc angle; and (2) at certain trench width and ridge width, the waveguide tends to have an extremely low leakage loss, which can be effectively referenced during waveguide etching.

Therefore, when utilizing the optoelectronic properties of lithium niobate waveguide, the leakage loss and transmittivity of lithium niobate ridge waveguide under different conditions can be utilized to select the most suitable ridge waveguide shape for design requirements and apply it in practice. This is very beneficial for us to study optical issues and utilize the polarization characteristics of lithium niobate. The next task is to explore how to use these rules to design efficient waveguide and perform precise fabrication.

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Declarations

Conflict of interest This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal's policies, and we believe that neither the manuscript nor the study violates any of these. There are no conflicts of interest to declare.

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